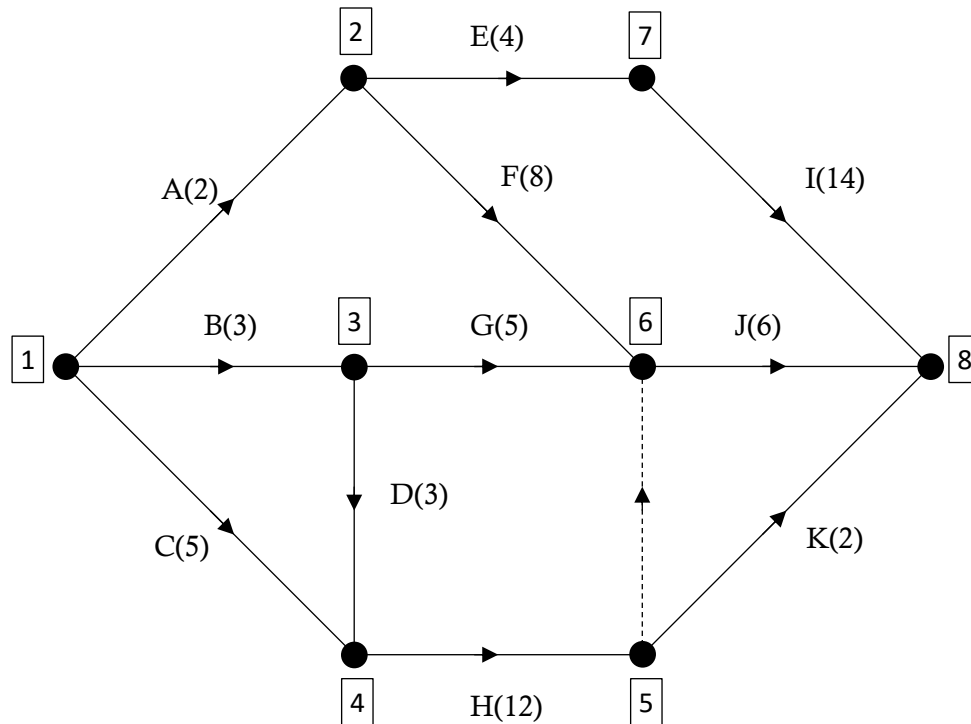


## ACTIVITY ON ARC DIAGRAMS

27/01/26

The following diagram is an activity-on-arc diagram for a task requiring 11 activities, labelled by A to K. The time required for each activity is shown in brackets next to the activity.



Question: What does this diagram show?

Question: Fill out the table below to show which prior tasks each activity depends on.

| Activity | Immediate Predecessors |
|----------|------------------------|
| A        |                        |
| B        |                        |
| C        |                        |
| D        |                        |
| E        |                        |
| F        |                        |
| G        |                        |
| H        |                        |
| I        |                        |
| J        |                        |
| K        |                        |

Question: Could activity H terminate at event 6, and the dummy activity go the 'other way' from event 6 to event 5 (after renumbering)?

### Exercise 1

Draw an activity-on-arc diagram for each of the following situations. Use the minimum number of dummy activities necessary in each case.

1.

| Activity | Description      | Immediate Predecessors | Time (minutes) |
|----------|------------------|------------------------|----------------|
| A        | Mix paint        | -                      | 5              |
| B        | Put up wallpaper | -                      | 120            |
| C        | Clean windows    | -                      | 50             |
| D        | Paint woodwork   | A                      | 30             |
| E        | Paint wallpaper  | A,B                    | 45             |

2.

| Activity | Description                   | Immediate Predecessors | Time (minutes) |
|----------|-------------------------------|------------------------|----------------|
| A        | Print out rules of croquet    | -                      | 6              |
| B        | Mow lawn                      | -                      | 25             |
| C        | Learn/practice croquet        | A                      | 30             |
| D        | Lay out croquet court on lawn | A,B                    | 13             |
| E        | Put away lawnmower            | B                      | 5              |
| F        | Set up drinks table on lawn   | B                      | 12             |

3.

| Activity | Description          | Immediate Predecessors | Time (minutes) |
|----------|----------------------|------------------------|----------------|
| A        | Chop vegetables      | -                      | 8              |
| B        | Get out pots         | -                      | 1              |
| C        | Boil kettle          | -                      | 4              |
| D        | Assemble salad       | A                      | 9              |
| E        | Fry vegetables       | A, B                   | 10             |
| F        | Cook pasta           | B, C                   | 12             |
| G        | Assemble adult meals | E, F                   | 3              |
| H        | Assemble child meal  | E, F                   | 1              |